

VSB ENGINEERING COLLEGE

(Affiliated to Anna University, Chennai)

Karur, Tamil Nadu

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

PROJECT REPORT ON

CYBERSHIELD AI: INTELLIGENT CYBER INCIDENT DETECTION AND SAFETY MANAGEMENT PORTAL

(SIH Problem Statement ID: SIH25183)

Submitted by

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VSB ENGINEERING COLLEGE

Karur, Tamil Nadu

Department of Electronics and Communication Engineering

BONAFIDE CERTIFICATE

This is to certify that the project report entitled "CyberShield AI: Intelligent Cyber Incident Detection and Safety Management Portal" is a bonafide record of the project work carried out by RATHISH R (Roll No: 922525106247) of II Year, Department of Electronics and Communication Engineering, VSB Engineering College, Karur, during the academic year 2025-2026, in partial fulfilment of the requirements for the Smart India Hackathon 2025 (Problem Statement ID: SIH25183).

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DECLARATION

I, RATHISH R (Roll No: 922525106247), hereby declare that the project report entitled "CyberShield AI: Intelligent Cyber Incident Detection and Safety Management Portal" submitted to VSB Engineering College, Karur, is a record of original work done by me under the guidance of Ms. AYSHA, Assistant Professor, Department of Electronics and Communication Engineering.

This project has not been submitted to any other university or institution for the award of any degree or diploma.

Place: Karur

Date: June 2026

RATHISH R

ABSTRACT

Cybercrime in India has escalated dramatically, with the Indian Computer Emergency Response Team (CERT-In) reporting over 13.9 lakh cybersecurity incidents in 2022 alone. The existing cyber incident reporting mechanisms, such as the National Cyber Crime Reporting Portal (cybercrime.gov.in), suffer from fragmented workflows, absence of intelligent threat analysis, and poor real-time communication with citizens. This project, developed in response to Smart India Hackathon 2025 Problem Statement SIH25183, presents CyberShield AI -- an intelligent, full-stack web portal that integrates Artificial Intelligence for automated cyber incident detection, risk scoring, and safety management.

The system is built using React 18 with TypeScript on the frontend, Supabase (PostgreSQL) as the Backend-as-a-Service with Row-Level Security, and Google Gemini AI for natural language processing and threat analysis. Key features include a citizen incident reporting dashboard, an AI-powered risk scoring engine, a real-time phishing link/email detector, an AI chatbot (CyberBot) for victim guidance, a public safety alert broadcasting system with audio notifications, and a secure administrative command centre with analytics. The application is deployed on Vercel and is accessible at cybersafe-india-portal.vercel.app.

Keywords: Cybersecurity, Artificial Intelligence, Incident Reporting, Phishing Detection, Gemini AI, React, Supabase, Smart India Hackathon.

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CHAPTER 1

INTRODUCTION

1.1 Background

The digital transformation of India has brought unprecedented connectivity, with over 900 million internet users as of 2024. While this digitisation has driven economic growth through platforms such as UPI, DigiLocker, and Aadhaar-based services, it has simultaneously exposed citizens to a growing landscape of cyber threats. According to the National Crime Records Bureau (NCRB), cybercrime cases registered in India increased by 11.8% in 2022 compared to the previous year. The Indian Computer Emergency Response Team (CERT-In) handled over 13.9 lakh cybersecurity incidents in 2022 alone, spanning phishing attacks, ransomware, website defacements, and distributed denial-of-service (DDoS) attacks.

The Government of India, recognising the criticality of cybersecurity, established the National Cyber Crime Reporting Portal (cybercrime.gov.in) and the helpline 1930 to enable citizens to report cyber incidents. However, the current system faces significant challenges: the reporting process is cumbersome, there is no real-time intelligent analysis of reported incidents, and the communication loop between authorities and citizens is slow. The Smart India Hackathon (SIH) 2025, organised by the Ministry of Education and AICTE, identified this gap through Problem Statement ID SIH25183, which calls for the development of an Intelligent Cyber Incident Detection and Safety Management Portal for the Army Cyber Group.

1.2 Problem Statement

SIH25183 -- Intelligent Cyber Incident Detection and Safety Management Portal. The problem statement requires the development of a centralised, intelligent platform capable of: receiving cyber incident reports from citizens securely; automatically analysing reported incidents using AI to determine risk levels and categorise threats; providing real-time safety alerts to the public; enabling authorities to monitor and respond to incidents efficiently; and offering AI-powered tools such as phishing detection and victim guidance chatbots.

1.3 Objectives

- To design and develop a secure, role-based web portal that enables citizens to report cyber incidents and administrators to manage them.
- To integrate Google Gemini AI for automated incident risk scoring, intelligent categorisation, phishing link/email analysis, and conversational victim support.
- To implement a real-time safety alert broadcasting system that enables authorities to issue advisories to citizens with audio notification support.
- To build an administrative analytics dashboard with visual representations of incident trends, severity distributions, and risk metrics.
- To deploy the application on a cloud platform (Vercel) for public accessibility.
- To implement Row-Level Security (RLS) at the database level to ensure strict data isolation between citizens and administrators.

1.4 Scope of the Project

The scope encompasses: User Management with Supabase Auth (email/password, role-based access); End-to-end Incident Lifecycle Management (submission, AI analysis, admin triage, resolution); Three AI capabilities powered by Google Gemini (risk scoring, phishing detection, chatbot); Safety Alert Broadcasting with Web Audio API notifications; Analytics dashboards using Recharts; and Production deployment on Vercel with GitHub CI/CD. The project does not include mobile native applications, integration with law enforcement databases (CCTNS), or real-time network intrusion detection.

CHAPTER 2

LITERATURE SURVEY

2.1 Review of Existing Technologies

2.1.1 National Cyber Crime Reporting Portal (cybercrime.gov.in)

The Government of India's official portal allows citizens to file complaints related to cybercrime, particularly financial fraud and crimes against women and children. The portal supports online FIR filing and tracks complaints through a reference number. Limitations: The portal uses a multi-page, text-heavy form that is not user-friendly. There is no AI-assisted analysis, no chatbot for guidance, and no mechanism for real-time safety alerts.

2.1.2 CERT-In Incident Reporting

The Indian Computer Emergency Response Team (CERT-In) accepts incident reports via email and a web form. It publishes advisories and vulnerability notes. Limitations: The reporting process is designed for IT professionals, not ordinary citizens. There is no automated triage, no risk scoring, and no public-facing dashboard.

2.1.3 International Systems

IC3 (FBI, USA) accepts cybercrime complaints via an online form but provides no real-time analysis or AI features. Action Fraud (UK) provides an online reporting tool with some automated categorisation but no AI chatbot. ACSC (Australia) provides a ReportCyber portal with structured forms but no AI integration.

2.2 Drawbacks of the Existing System

S.No.	Drawback	Impact
1	No AI risk assessment	Incidents need manual review
2	No phishing tool	Citizens cannot verify links
3	No AI chatbot	No immediate advice
4	No real-time alerts	Cannot warn citizens fast
5	Poor UI/UX	Complex forms stop reporting
6	No analytics	Decision-makers lack insights
7	No DB security	Data isolation uses app logic

CHAPTER 3

PROPOSED SYSTEM

3.1 System Overview

The proposed system, CyberShield AI, is a full-stack web application designed as a dual-portal system:

Citizen Portal: A user-friendly dashboard where registered citizens can report cyber incidents through an intuitive form, receive AI-generated risk scores immediately, track report status, interact with CyberBot (AI chatbot) for guidance, use the AI Phishing Detector, and view public safety alerts.

Admin Command Centre: A secure, restricted dashboard where administrators can view all incidents with search/filter capabilities, update statuses, view full details including AI analysis, access visual analytics, and create/manage public safety alerts.

The system uses Google Gemini AI (gemini-2.5-flash) for three distinct AI capabilities. The backend is powered by Supabase (PostgreSQL with Auth and RLS). The frontend is built with React 18 and TypeScript, styled with Tailwind CSS, and deployed to Vercel.

3.2 Advantages over Existing System

Feature	Existing System	CyberShield AI
Reporting Form	Multi-page forms	Single-page form
Risk Assessment	None	AI scoring (0-100)
Categorisation	Manual	AI-suggested type
Phishing tool	None	Gemini AI analyser
Victim Guidance	Static FAQ pages	AI chatbot
Safety Alerts	Static notices	Broadcast + audio
Admin Analytics	None	Interactive charts
Access Control	App-level only	Database-level RLS
UI/UX	Government design	Modern cyber theme

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

The system follows a three-tier client-server architecture with Presentation Layer (React 18 + TypeScript + Tailwind CSS), Application Layer (AuthContext, AI Services via Gemini API, Supabase Client with REST + RLS), and Data Layer (Supabase PostgreSQL + Auth + RLS, Google Gemini API, Vercel CDN & Hosting).

```
PRESENTATION LAYER
|-- Citizen Dashboard (Report, My Incidents, CyberBot)
|-- Admin Command Centre (Dashboard, Incidents, Analytics)
|-- Public Pages (Landing, Alerts, Login, Signup)
    |
APPLICATION LAYER
|-- AuthContext (Session Management)
|-- AI Services (Gemini API)
|-- Supabase Client (REST)
    |
DATA LAYER
|-- Supabase PostgreSQL + RLS
|-- Google Gemini API
|-- Vercel CDN & Hosting
```

4.2 Module-wise Breakdown

The application is divided into 8 functional modules:

Module 1: Authentication & Management

User registration, Supabase Auth, automatic profile creation, and role-based route protection.

Module 2: Citizen Incident Reporting

Guided form capturing title, type, severity, description, phone, platform, date, financial loss. AI scoring on submission.

Module 3: AI Risk Scoring Engine

Client-side keyword scoring + Gemini AI comprehensive analysis with reasoning.

Module 4: AI Chatbot (CyberBot)

Floating widget with Gemini multi-turn conversation, custom instructions for India, quick buttons.

Module 5: AI Phishing Detector

URL/email analysis via Gemini AI returning verdict, threat score, red flags, and explanation.

Module 6: Admin Command Centre

Dashboard with KPI cards, searchable incident table with detail modals, visual analytics.

Module 7: Safety Alert Broadcasting

Admin CRUD for alerts. Citizen page with synthesised audio notifications (Web Audio API).

Module 8: Dashboard Layout

Responsive sidebar with role-adaptive links, user profile display, and logout.

4.3 Database Design

Table	Key Columns	Purpose
profiles	id, email, full_name, role	User profile data
incidents	id, title, type, severity	Cyber incident reports
alerts	id, title, content, severity	Public safety alerts
ai_analyses	id, risk_score, keywords	AI analysis results

4.4 Technology Stack

Layer	Technology	Purpose
Frontend	React 18.3 + TS 5.5	Component-based UI
Build Tool	Vite 5.4	Fast dev server
CSS	Tailwind CSS 3.4	Utility-first styling
Routing	React Router 7.17	Client navigation
Charts	Recharts 3.8	Data visualisation
AI Engine	Gemini (2.5-flash)	NLP, risk analysis
Backend	Supabase (Postgres)	DB, Auth, RLS
Deployment	Vercel	CDN hosting

CHAPTER 5

IMPLEMENTATION

5.1 Development Setup

The project was initialised using the Vite build tool with the React-TypeScript template:

```
npx create-vite@latest cybersafe-portal --template react-ts  
npm install @supabase/supabase-js @google/generative-ai  
npm install -D tailwindcss
```

5.2 Authentication Module

Implemented using Supabase Auth with a Context Provider pattern. ProtectedRoute.tsx enforces role-based access, redirecting citizens from admin pages.

5.3 Incident Reporting

The interactive single-page form collects structured data with fields for title, category, severity level, description, platform, date, and financial loss. The AI Risk Scoring Engine processes data before inserting into the database.

5.4 AI Risk Scoring Engine

A dual-layer system: Layer 1 uses keyword-based scoring (bank, password, etc.). Layer 2 sends the incident to Google Gemini for comprehensive analysis returning a risk score and reasoning.

5.5 AI Chatbot (CyberBot)

A floating widget communicating with Google Gemini using multi-turn conversation. Features include India-specific guidance, conversation history, and a model fallback chain.

5.6 Admin Command Centre

Dashboard with real-time KPI cards. AdminIncidents.tsx provides a searchable table with clickable rows opening detail modals showing full AI analysis. AdminAnalytics.tsx provides charts for trends, severity, and categories.

5.7 Safety Alerts

Administrators create alerts. Citizens view them with a synthesised notification sound using the Web Audio API with an exponential frequency sweep.

5.8 Challenges & Solutions

Challenge	Solution
Silent database failures	Explicit error checking added
Gemini API rate limiting	Model fallback chain + retry delay
DB schema mismatches	ALTER TABLE statements added
Hardcoded risk scores	Supabase RPC function for dynamic data

CHAPTER 6

RESULTS AND DISCUSSION

6.1 System Outputs

The CyberShield AI portal was successfully deployed to production. Key outputs include:

- Landing Page: Cyberpunk-themed interface with live average risk score.
- Citizen Dashboard: Role-appropriate navigation with quick-access cards.
- Incident Form: Severity selector with colour-coded feedback.
- CyberBot Chatbot: Floating widget responding to queries with context.
- Phishing Detector: Colour-coded verdict cards with detailed red flags.
- Admin Analytics: Three chart types with configurable time ranges.

6.2 Performance Metrics

Metric	Observed Value
Page Load	< 2 seconds (Vercel)
Supabase Query	50-150 ms
Gemini API Time	1-3 seconds
Lighthouse Score	85+ Performance, 90+ Accessibility
RLS Enforcement	100% strict data isolation

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

The CyberShield AI portal was successfully designed and deployed for Smart India Hackathon 2025. It demonstrates that integrating Artificial Intelligence into cyber incident management improves efficiency and accessibility.

- Production-deployed web portal built with React, Supabase, and Tailwind.
- Three AI-powered features via Google Gemini (scoring, phishing, chatbot).
- Secure, role-based system with database-level RLS.

7.2 Future Enhancements

- Mobile Application: Native Android/iOS apps for broader accessibility.
- Integration with CCTNS: Direct API integration for incident escalation.
- Multi-language Support: Localisation into regional Indian languages.
- File Upload: Cloud-based evidence attachment using Supabase Storage.
- Push Notifications: Browser push and SMS alerts using Twilio.

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END OF REPORT

*CyberShield AI: Intelligent Cyber Incident Detection
and Safety Management Portal*